Xu Pan M.Sc.

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Experiences

Nanyang Technological University (NTU) Singapore | 2026.08 - present
Research Associate | *School of Electrical and Electronic Engineering, College of Engineering*

- Under the supervision of Prof. Ziwei Wang, at Perception and Embodied Intelligence (PINE) Lab
- VLA Reinforcement Learning: Investigating policy optimization and exploration for flow-based VLAs.
- Dual-System Architectures: Exploring asynchronous VLAs with learned latent-state updates for efficient control.
- Adaptive Robot Learning: Developing feedback-conditioned in-context learning and flow guidance for manipulation.

Agency for Science, Technology and Research (A*STAR) Singapore | 2025.08 - 2026.06
Research Assistant | *Centre for Frontier AI Research (CFAR), Institute of High Performance Computing (IHPC)*

- Pioneered spatially-aware reinforcement learning adaptation for flow-based VLA models, preserving implicit spatial inductive bias through fused geometric prior and preventing degradation under distribution shifts.
- Designed geometry-aligned dense rewards and spatially-conditioned exploration, enabling stable RL fine-tuning, superior zero-shot spatial generalization, and robust transferable behaviors in complex manipulation scenes.

Baidu International Technology (Shenzhen) Co., Ltd. Shenzhen, China | 2024.12 - 2025.03
Research Intern | *Foundation Public Technology Group, International Technology R&D Department*

- Developed DiT-based framework for controllable, arbitrarily long video generation via key-frame conditioning.
- Improved the visual fidelity of rendered text in image generation by designing specialized attention mechanisms.
- Contributed to the **SynClub** AI chat app through assessment of frontier multimodal generative models and optimization of text-to-image/video modules, delivering improved performance and higher-quality outputs.

Education

Wuhan University | **State Key Lab. LIESMARS** Wuhan, China | 2023.09 - 2026.06
M.Sc. in Cartography and Geographic Information Systems

- GPA: 3.54/4.00, Average Score: 89.40/100, IELTS: 7.0(6.0) | Research area: Computer Vision and 3D Reconstruction

Wuhan University | **School of Remote Sensing and Information Engineering** Wuhan, China | 2019.09 - 2023.06
B.Eng. in Remote Sensing Science and Technology

- GPA: 3.75/4.00, Average Score: 88.90/100, Recommended Exemption

Publications

1. **Xu Pan**, Zhenglin Wan, Xingrui Yu, Xianwei Zheng, Youkai Ke, Ming Sun, Rui Wang, Ziwei Wang, Ivor Tsang. **SA-VLA: Spatially-Aware Reinforcement Learning for Flow-Matching Vision-Language-Action Models**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshop (CVPRW) on 3D-LLM/VLA*, 2026. Available at `https://arxiv.org/abs/2602.00743`
2. **Xu Pan**, Zimin Xia, Xianwei Zheng. **Scale-aware Co-visible Region Detection for Image Matching**. *ISPRS Journal of Photogrammetry and Remote Sensing [IF 12.2]*, 2025. `https://doi.org/10.1016/j.isprsjprs.2025.08.015`
3. **Xu Pan**, Zhen Pang, Qiyuan Ma, Mingyue Dong, Wei Ji, Xianwei Zheng. **SAMatcher: Dense Co-Visibility Modeling via Cross-View Fusion for Scale-Imbalance Image Matching**. *Under Review at IEEE Transactions on Circuits and Systems for Video Technology*, 2026. Available at `https://arxiv.org/abs/2606.03406`
4. Ming Sun, Rui Wang, Xingrui Yu, Lihua Jing, Hangyu Du, Zhenglin Wan, **Xu Pan**, Ivor Tsang. **Towards Backdoor-Based Ownership Verification for Vision-Language-Action Models**. *Under Review at IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2026. Available at `https://arxiv.org/abs/2605.09005`
5. Jiashen Huang, Yu Jia, **Xu Pan**. **Trusting AI in China: Institutional Trust and the Domestic Advantage of DeepSeek over ChatGPT**. *Under Review*, 2026. Available at `https://arxiv.org/abs/2606.01228`.

Projects

Intelligent Interpretation Platform for Remote Sensing Images Based on Baidu PaddlePaddle 2022.03 - 2022.08
`TwSphinx54/I2RSI`

Web-based Global Natural Disaster Assessment System 2022.05 - 2022.09
`TwSphinx54/GNDAS`

Academic Activities

2026 Vision and Learning Seminar (VALSE), China Society of Image and Graphics (CSIG) | Wuhan, China 2026.05.08 - 05.10

2025 Annual Conference on Photogrammetry and Remote Sensing, Chinese Society for Geodesy, Photogrammetry and Cartography (CSGPC) | Kunming, China 2025.07.28 - 07.30

Awards and Honors

- **Outstanding Graduating Graduate Student**, Wuhan University 2026.05
- **Outstanding Graduate Student**, Wuhan University 2025.11
- **Graduate Academic Excellence Scholarship**, Wuhan University 2025.11
- **Graduate Academic Excellence Scholarship**, Wuhan University 2024.11
- **Outstanding Student Leader**, Wuhan University 2024.10
- **Outstanding Student Club Leader**, Wuhan University 2024.08
- **Active Contributor to Social Activities**, School of Remote Sensing and Information Engineering 2023.05
- **Wuhan University Class C Scholarship**, Wuhan University 2022.10
- **National Second Prize**, China Software Cup College Student Software Design Competition 2022.08
- **Honorable Mention**, Mathematical Contest In Modeling 2022.03
- **Third Prize**, Asia and Pacific Mathematical Contest in Modeling 2021.12
- **First Prize in Hubei Division**, China Undergraduate Mathematical Contest in Modeling 2021.10
- **Outstanding Student**, Wuhan University 2021.09
- **Bronze Medal**, China Collegiate Algorithm Design & Programming Challenge Contest 2021.03
- **Second Prize in Final**, Translation & Interpreting Contest of Hubei Province 2020.05

Skills

Languages: Mandarin, English (IELTS 7.0)

Programming: `Python`, `C++`, `MATLAB`, `HTML`, `CSS`, `JavaScript`, `C#`, `SQL`, `R`

Soft Skills: Leadership, Project Management, Event Organization, Public Speaking, Design & Editing

Certificates

System Integration Project Management Engineer
Qualification Certificate of Computer and Software Technology Proficiency, MOHRSS & MIIT

Certificate of Level 4 in Network Engineer
National Computer Rank Examination, NEEA

Certificate of Level 3 in Network Technology
National Computer Rank Examination, NEEA

Certificate of Level 3 in Database Technology
National Computer Rank Examination, NEEA

Academic Services

Reviewer 2027 IEEE International Conference on Robotics & Automation (ICRA'27)
Reviewer 2027 International Joint Conference on Neural Networks (IJCNN'27)
Reviewer 10th Annual Conference on Robot Learning (CoRL'26)
Member ISPRS Student Consortium (ISPRS SC)
Volunteer 2022/2023 International Graduate Workshop on Geo-Informatics

Leadership

Chairman The 22nd LIESMARS Graduate Student Union
Director GeoScience Café Operation Center